

Project ABEONA Crew Capsule Main Engine: Thrust Chamber, Feed System, and Pressurization Design

[Your Name]

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1 Introduction

This analysis designed the single 25 kN main engine of the Project ABEONA Crew Capsule service module and its propellant feed and pressurization system. The engine burns nitrogen tetroxide (NTO) and monomethylhydrazine (MMH), is pressure-fed by regulated helium, and operates only in vacuum. The work covered the inputs, the cycle and operating-point trades, the thrust chamber, the vacuum nozzle, the thermal design, the injector, the tanks and pressurization, the materials, and consistency checks.

All vehicle numbers were taken directly from the Crew Capsule tab of `Mission_Breakdown.xlsx` [12]. The Preliminary Design Report (PDR) was used for the propellant choice, architecture and packaging envelope [12]. The method and reporting style follow the Stage 2/3 hydrolox engine report [13] and Huzel and Huang [1]. Every number in this report was produced by the Python analysis scripts.

NASA CEA could not be installed in the analysis environment. Combustion performance was computed with an equilibrium solver written with the CEA method (NASA RP-1311 [2]) and the NASA thermodynamic database that ships with CEA [11]. It matched CEA output within 0.02 % (Table 7). Liquid MMH and NTO properties came from the RocketProps library [6].

2 Requirements and Inputs

The spreadsheet values used in the design are listed in Table 1; they were used as given. The mass flow and burn time in the sheet correspond to the minimum 280 s I_{sp} ; the designed engine has a higher I_{sp} , so at the same 25 kN it uses less propellant per second and fires longer (Sec. 11).

Table 1: Crew Capsule requirements (spreadsheet).

Parameter	Spreadsheet value
Initial / propellant / structural / payload mass, kg	38,000 / 21,000 / 6,500 / 10,500
Burnout mass, kg; mass ratio	17,000; 2.235294
ϵ / λ	0.236364 / 0.381818
Required I_{sp} , s; u_{eq} , m/s	280; 2,746.8
Δu , m/s	2,209.45
Mass flow, kg/s; burn time, s	9.1015; 2,307.3
Thrust, engines	25 kN, 1
T/W Earth, initial / burnout	0.0671 / 0.1499
T/W Moon, initial / burnout	0.405 / 0.905
MMH / NTO density, kg/m ³	875 / 1,440

Differences between the spreadsheet and the PDR are listed in Table 2; the spreadsheet value was carried in every case. The PDR tanks (2.6 m spheres, 9.203 m³) leave only 1.4 % (NTO) and 1.6 % (MMH) free volume for the spreadsheet propellant load, which is not enough for ullage gas or a propellant management device, so the tanks were resized (Sec. 9.2).

Table 2: Inputs audit: spreadsheet versus PDR.

Item	Spreadsheet	PDR	Disposition
Payload ratio λ	0.3818	0.304	Spreadsheet carried
Δu , m/s	2,209.5	2,200	Spreadsheet carried
Mixture ratio	not given	1.65 (AJ10)	PDR value used
Launch Escape System	Stage 3 payload 45,000 kg	7,000 kg (Fig. 9)	Outside the 38,000 kg capsule mass
Propellant volume, m ³	NTO 9.080, MMH 9.057	9.203 each	Only 1.4 / 1.6 % free; tanks resized
Aft envelope	–	2.01 m base to exit; 0.5 m cone	Engine length is the limit

3 Design Decisions and Assumptions

Table 3: Approved design decisions.

Item	Decision
Propellants	NTO / MMH
Cycle	Pressure-fed, regulated helium
Chamber pressure	0.862 MPa (125 psia), nozzle-entrance stagnation pressure (AJ10-190 heritage)
Mixture ratio	1.65 overall (PDR)
Nozzle	Rao optimum-thrust bell for vacuum, 80 % length, area ratio set by the engine-length envelope
Cooling	Ablative silica-phenolic chamber and throat; fuel-rich outer injector ring; C-103 nozzle extension
Pressurization	Helium in composite bottles, two series regulators, check valves, relief valve
Propellant acquisition	Screen-channel propellant management device (PMD) in each tank

Main assumptions:

1. Steady state; combustion products are ideal gases in chemical equilibrium; $g = 9.81 \text{ m/s}^2$ as in the spreadsheet.
2. Delivered efficiency calibrated to the AJ10-190 Orbital Maneuvering Engine (OME): 315.1 s at 0.862 MPa, area ratio 55, O/F 1.65 [15,16]. Its ideal value is 334.6 s, so $\eta_{I_{sp}} = 0.9417$. The pessimistic case uses the R-4D-11 (310 s against 348.0 s ideal, $\eta_{I_{sp}} = 0.8908$) [15]. $\eta_{c^*} = 0.99$ [16].
3. Propellants at 293 K; tank volumes use the spreadsheet densities.
4. Gas-side heat transfer by Bartz [9] with no hypergolic calibration; a 1.3 factor was used to size the wall protection.
5. Ablator properties, material limits, component pressure drops and some component masses are assumed values, to be verified.

4 Cycle and Operating-Point Selection

4.1 Envelope

In PDR Fig. 9 the service-module section is 6 m long and 4.98 m in diameter. The 2.01 m dimension runs from the service-module base to the nozzle exit, and the 0.5 m dimension is the aft-cone height. The resized tanks leave 0.63 m free in the 6 m section below the NTO tank, so the engine (gimbal plane to nozzle exit) may be 2.642 m long. The length, not the 2.0 m drawn exit diameter, limited the nozzle at every chamber pressure studied.

4.2 Cycle

Five feed cycles were sized with the same mass models (Table 4). A blowdown system needs 50 % more tank volume and its thrust falls 3:1 as the tanks empty. Regulating first and blowing down at the end saves only a few kilograms of helium. Pump-fed engines reach higher pressure and area ratio, but the gas-generator figures come from an uprated OME that never flew [16] and the electric pump needs a large battery. The pressure-fed regulated cycle was selected: it is the AJ10-190 / European Service Module arrangement, it has no rotating machinery, and it restarts simply.

Table 4: Cycle trade (nominal efficiency).

Cycle	P_c , MPa	ε	I_{sp} , s	System mass, kg	Remark
Pressure-fed, regulated He	0.86	106	321.7	1,121	heritage AJ10-190 / ESM
Pressure-fed, blowdown 3:1	2.59	106	321.7	2,937	tank volume 27.3 m ³ (x1.50), thrust 75 - 25 kN
Regulated then blowdown (last 15 Electric pump-fed	2.50	295	329.9	1,000	45 kW, battery 357 kg, pumps 50 kg
Gas-generator pump-fed	2.41	285	326.4	663	UOME (never flown)

4.3 Chamber pressure

Raising the chamber pressure shrinks the throat, so a larger area ratio fits the same length and I_{sp} rises. However the tanks must run at about twice the chamber pressure, so tank, helium and bottle mass grow quickly (Table 5, Fig. 1). The lightest system is near 0.6 MPa. The OME value of 0.862 MPa was selected because the efficiency calibration applies there directly and the system is still light.

Table 5: Pressure-fed chamber-pressure sweep.

P_c , MPa	ε	D_e , m	I_{sp} , s	p_{tank} , MPa	Tanks, kg	He, kg	Bottles, kg	Engine, kg
0.600	75.4	1.484	318.3	1.20	443	41	192	143
0.862	106.2	1.462	321.7	1.72	637	60	281	143
1.000	122.3	1.455	323.0	2.00	739	71	330	143
1.250	151.3	1.444	324.8	2.50	923	90	420	142
1.500	180.2	1.436	326.2	3.00	1,108	110	515	142
2.000	237.9	1.426	328.3	4.00	1,477	153	717	142
2.500	295.4	1.418	329.9	5.00	1,847	200	936	141

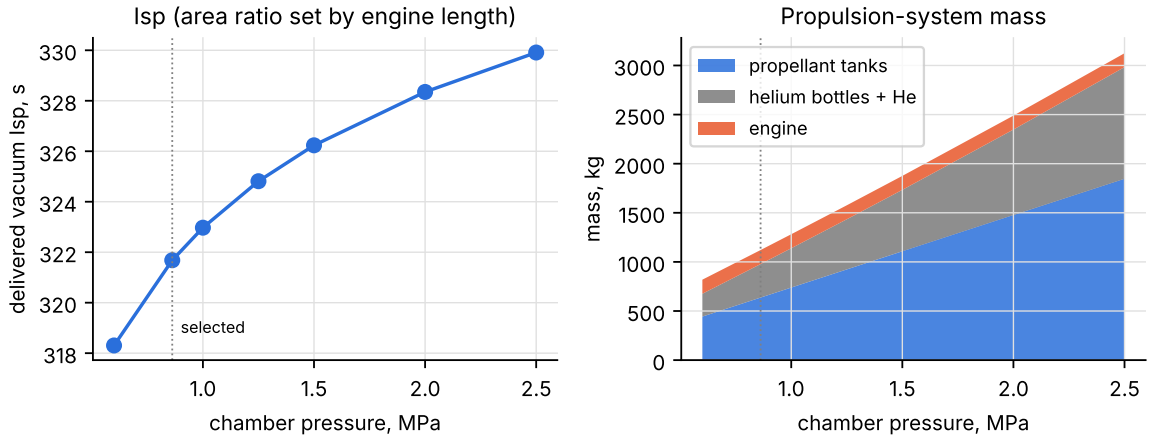


Figure 1: Chamber-pressure trade.

4.4 Mixture ratio

The ideal I_{sp} peaks at O/F 2.40 with shifting equilibrium and 1.70 with frozen chemistry (Fig. 2); a real engine of this size peaks between the two, near 1.9 [5]. The equal-volume O/F with the spreadsheet densities is 1.646, which is why the PDR tanks are equal. O/F 1.65 was kept. The core of the injector runs at O/F 1.92 anyway because the outer ring is fuel-rich (Sec. 7).

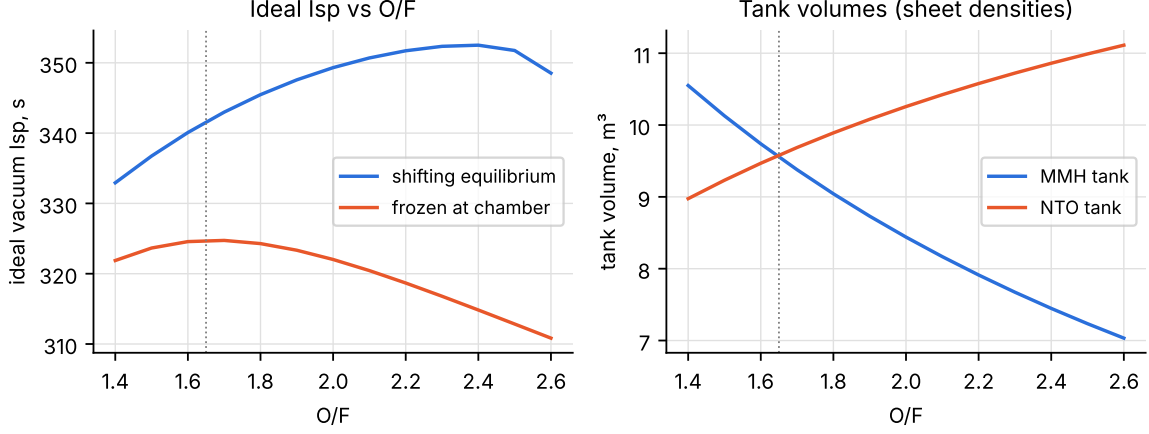


Figure 2: Ideal I_{sp} and tank volumes versus O/F.

5 Thrust Chamber

5.1 Ideal performance

Table 6 gives the ideal vacuum performance at 0.862 MPa, O/F 1.65 and area ratio 108. Reactant heats of formation were the CEA input-card values (NTO -19.56 kJ/mol, MMH $+53.97$ kJ/mol); the database values give 332.8 s instead of 332.4 s at the validation point. A closed-form one-dimensional nozzle with the chamber γ reproduced the frozen c^* within 0.07 %.

Table 6: Ideal vacuum performance.

Quantity	Shifting	Frozen	Closed form, diff. vs frozen
T_c , K	3,042.3	3,042.3	—
c^* , m/s	1,737.9	1,698.6	1,699.7 (+0.07 %)
$C_{F,vac}$	1.9276	1.8753	1.9143 (+2.08 %)
$I_{sp,vac}$, s	341.5	324.7	331.7 (+2.15 %)
Exit pressure, Pa	430	331	—

Table 7: Solver validation against NASA CEA.

Quantity	NASA CEA	This solver	Diff., %
T_c , K	3,466.8	3,466.8	-0.001
c^* shifting, m/s	2,350.6	2,350.6	+0.001
C_F shifting	1.9739	1.9738	-0.006
I_{sp} shifting, s	473.1	473.1	+0.002
c^* frozen, m/s	2,319.6	2,319.6	-0.002
C_F frozen	1.9329	1.9328	-0.006
I_{sp} frozen, s	457.2	457.2	-0.008
NTO/MMH, O/F 1.0–2.9: largest difference	–	–	I_{sp} 0.02, c^* 0.00, T_c 0.00

5.2 Delivered performance and geometry

The model reproduces the OME flight value at its own area ratio (315.1 s against 315.1 s). The delivered vacuum I_{sp} of this engine is 320.8 s (303.5 s pessimistic), 41 s above the 280 s requirement. The geometry is in Table 8.

Table 8: Thrust-chamber geometry and delivered performance.

Parameter	Value	Basis
Mass flow (NTO / MMH), kg/s	7.9434 (4.9459 / 2.9975)	$F/(I_{sp}g)$
Delivered vacuum I_{sp} (pessimistic), s	320.8 (303.5)	AJ10-190 (R-4D-11) calibration
Delivered c^* , $C_{F,vac}$	1,675.1 m/s, 1.8789	η_{c^*} 0.99
Throat area / diameter	154.36 cm ² / 140.2 mm	$\dot{m}c^*/P_c$
Area ratio / exit diameter	108 / 1.457 m	engine-length limit
Contraction ratio / chamber diameter	3.0 / 242.8 mm	H&H practice (assumed)
L^* / chamber volume	0.80 m / 12.35 L	H&H Table 4-1 (assumed)
Cylinder / convergent length	182 / 136 mm	30° cone, $R_1 = 1.5r_t$
Injector-end pressure	0.8827 MPa (1.0241 P_c)	Rayleigh
Injector face to throat / throat to exit	0.318 / 1.968 m	80 % bell
Engine length incl. 0.35 m head	2.636 m	available 2.642 m

6 Vacuum Nozzle

6.1 Contour

The contour was designed with an axisymmetric method-of-characteristics (MOC) solver. Along the characteristics

$$d(\theta \pm \nu) = \pm \frac{\sin \theta \sin \mu}{y} ds, \quad (1)$$

(upper sign on C^-) [7,8]. The wall is a $0.382r_t$ arc from the throat to θ_N followed by a smooth curve to the exit lip at angle θ_E . θ_N and θ_E were searched for maximum thrust at fixed length and exit diameter, which is Rao’s optimum-thrust problem [10]. The solver was checked against a 15° cone (0.98397 against $(1 + \cos \alpha)/2 = 0.98296$) and a flat wedge (0.98882 against $\sin \alpha/\alpha = 0.98862$), with 0.11 % mass-flow error. Results are in Table 9 and Fig. 3.

Table 9: Rao vacuum bells at area ratio 108 (MOC).

Length (% of 15° cone)	70	80	90	100	cone
Length, m	1.722	1.968	2.214	2.461	2.461
θ_N , deg	39.0	37.0	34.0	34.0	15.0
θ_E , deg	9.0	8.0	7.0	4.0	15.0
Divergence efficiency	0.98225	0.98854	0.99201	0.99414	0.98397

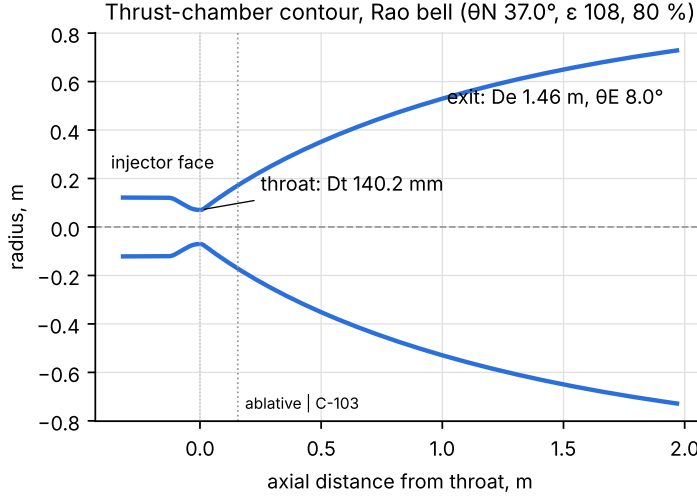


Figure 3: Thrust-chamber contour from the injector face to the nozzle exit.

6.2 Vacuum-optimization check

The engine fires only in space, so it was checked three ways that the nozzle is optimized for vacuum:

1. The MOC thrust integral uses zero outside pressure, so the bell maximizes vacuum thrust.
2. The exit pressure is only 430 Pa (0.42 % of sea-level pressure), with a vacuum thrust coefficient of 1.879. At sea level the flow would separate inside the bell, so this is a vacuum-only nozzle, as intended.
3. The length is fixed by the service module, so a shorter bell allows a larger area ratio. Table 10 and Fig. 4 show the delivered vacuum I_{sp} for each bell length at the same overall engine length. The best is the 70 % bell, 0.10 s above the selected 80 % bell, which is within the accuracy of the method. The 80 % bell was kept.

Table 10: Vacuum I_{sp} at the same engine length.

Bell length, %	Area ratio	Exit diameter, m	Divergence eff.	Vacuum I_{sp} , s
70	139	1.650	0.9823	320.97
80	109	1.461	0.9885	320.88
90	87	1.314	0.9920	319.98
100	72	1.197	0.9941	318.76

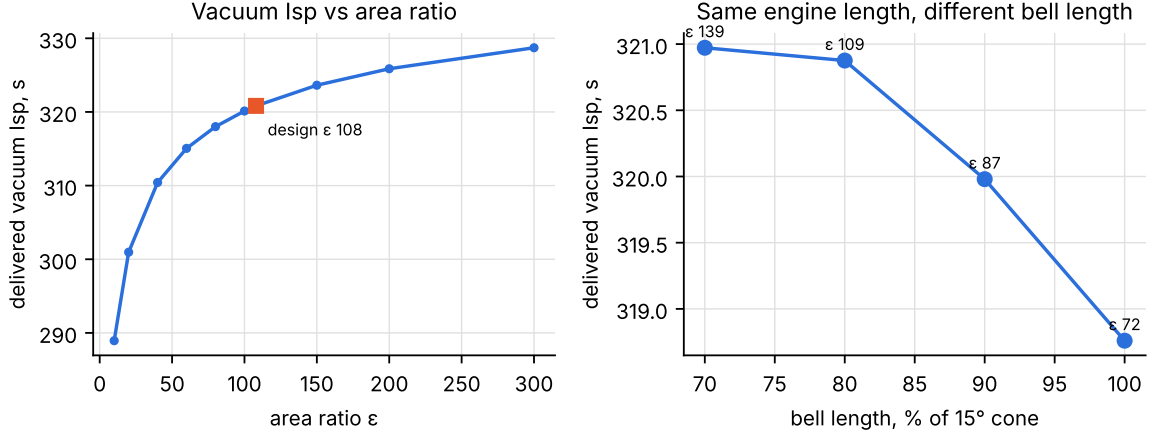


Figure 4: Left: delivered vacuum I_{sp} against area ratio. Right: bell length at fixed engine length.

7 Thermal Design

7.1 Heat flux

The gas-side heat transfer coefficient followed Bartz [9; 1, Eq. 4-13]:

$$h_g = C \frac{0.026}{D_t^{0.2}} \left(\frac{\mu^{0.2} c_p}{P_r^{0.6}} \right)_0 \left(\frac{P_c}{c^*} \right)^{0.8} \left(\frac{D_t}{R_c} \right)^{0.1} \left(\frac{A_t}{A} \right)^{0.9} \sigma. \quad (2)$$

At the throat h_g is 2,286 W/(m²K). The heat flux along the engine is plotted in Fig. 5; it peaks at the throat at 3.1 MW/m² into a 700 K wall.

7.2 Fuel-rich outer ring

The combustion gas is about 3,000 K, well above what any ablative can survive for the 2,644 s of total firing. The injector's outer ring of elements was therefore run fuel-rich, at O/F 0.6. This puts a layer of cooler (1,541 K), fuel-rich gas against the wall, much as the OME uses a fuel film. 12.5 % of the flow goes through the outer ring. That is the smallest amount that keeps the throat surface below the temperature at which silica phenolic starts to wear away, even with the heat flux raised by 30 %. It costs 0.78 s of I_{sp} , which is already included in the delivered value. A fuel film from separate orifices was the alternative; the fuel-rich ring was chosen because it uses the same injector elements and stays attached to the wall over the whole chamber.

7.3 Ablative liner and nozzle extension

Material choice. The chamber and throat are lined with tape-wrapped silica phenolic. The two common rocket ablatives are silica phenolic and carbon phenolic. Carbon phenolic handles higher temperatures, but its carbon reacts with the water vapor and carbon dioxide in NTO/MMH exhaust and wears away, and it conducts heat well, so it needs a thicker insulating layer behind it. Silica phenolic forms a glassy surface that holds up in this exhaust, insulates well, and is the material used on the Apollo service-module engine and the Delta second-stage AJ10 engines. Silica phenolic was chosen.

How it works. As the liner heats, the resin near the hot surface turns to char, which insulates the material behind it. The liner wears away only if its surface passes about 1,900 K. The fuel-rich ring keeps the hottest point, the throat, at 1,806 K, so the throat does not grow during the mission.

Results. The liner runs from the injector face to area ratio 6. It was sized for 3,305 s of firing, 25 % more than the total firing time. The char reaches 51 mm deep, and the liner is 83 mm thick (66 kg) so

that the outer titanium case stays below 420 K; it ends at 354 K (Fig. 7). Beyond area ratio 6 the gas is cool enough for a thin C-103 niobium skirt with a silicide coating, as on the Apollo service-module engine; it peaks at 1,389 K, under its 1,600 K limit (Fig. 6). After the longest firing, heat stored in the liner flows back into the injector; an insulating ring between the two keeps the injector near 399 K, below the 442 K boiling point of MMH in the manifold. Without it the injector would rise about 152 K.

Table 11: Thermal design results.

Item	Value
Liner material	silica phenolic (tape wrapped)
Liner covers	injector face to area ratio 6 (0.474 m long)
Fuel-rich outer ring	12.5 % of the flow at O/F 0.6 (1,541 K gas)
Hottest liner surface (throat)	1,806 K (wear starts at 1,900 K)
Liner thickness / mass	82.6 mm / 66.4 kg
Charred depth at end of firing	50.9 mm (design life 3,305 s)
Outer case temperature at end of firing	354 K (limit 420 K)
Surface wear (recession)	1.38 mm max., none at the throat
Nozzle extension	C-103 from area ratio 6, max. 1,389 K
Injector after a long firing	about 399 K with insulating ring

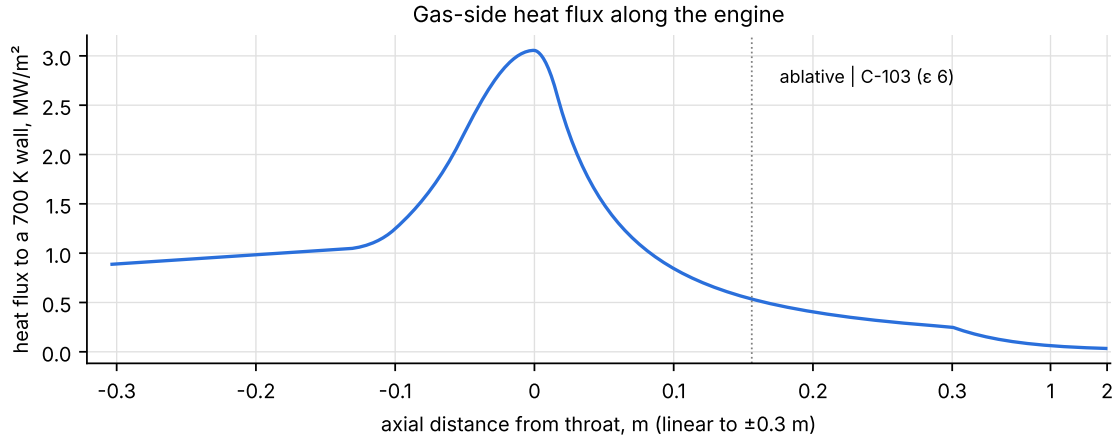


Figure 5: Gas-side heat flux along the engine.

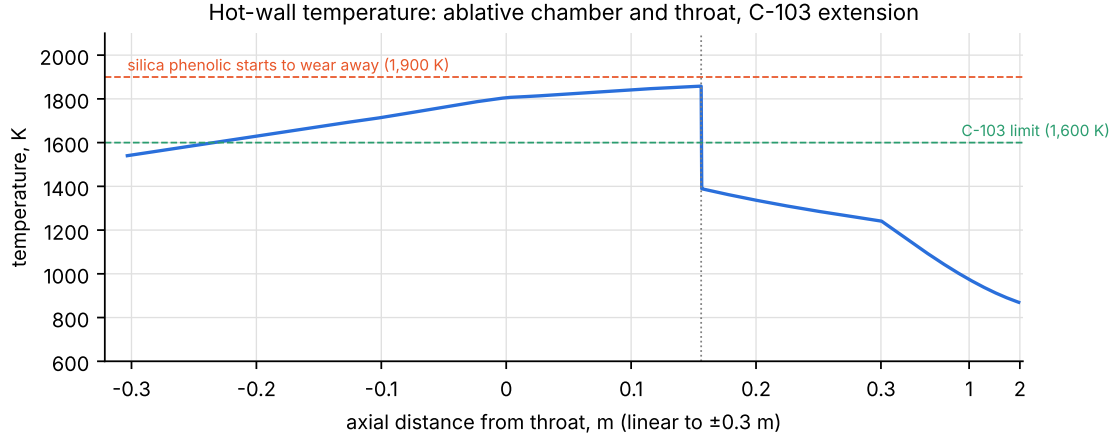


Figure 6: Hot-wall temperature: silica-phenolic liner (left of the dotted line) and C-103 extension.

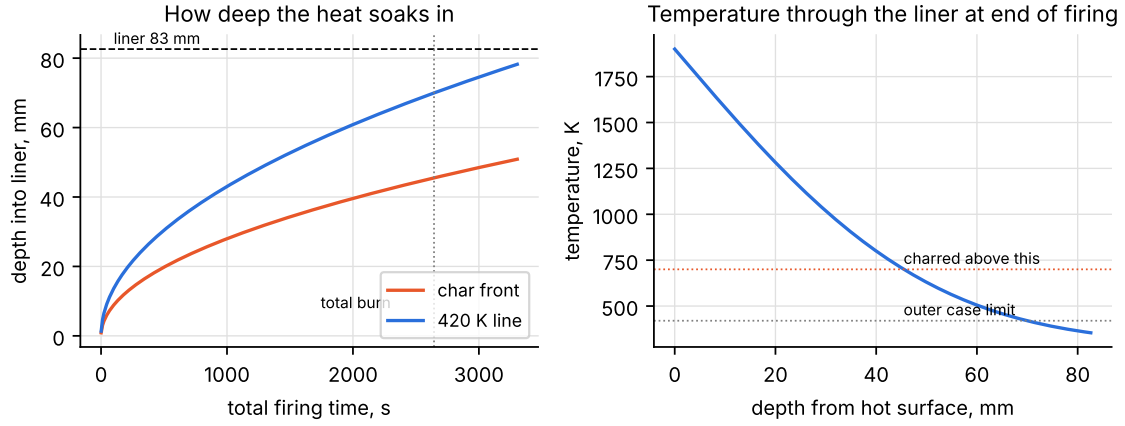


Figure 7: Silica-phenolic liner at the throat.

8 Injector

Unlike-doublet (NTO–MMH) impinging elements were used, as on the Apollo service-module engine. The NTO pressure drop is 20 % of the injector-end pressure [1]; the MMH orifices were sized for the best mixing by Rupe’s criterion $\rho_o V_o^2 d_o = \rho_f V_f^2 d_f$, giving 26 %. Both drops exceed 15 %, which keeps the feed system from coupling with the combustion. Twelve quarter-wave acoustic cavities, as on the OME [16], are tuned to the first tangential mode and are 59 mm deep (Tables 12 and 13).

Table 12: Injector design.

Parameter	Core	Outer ring
Element type / count	unlike doublet / 120	unlike doublet / 48
NTO / MMH per element, g/s	38.1 / 19.8	7.8 / 12.9
Orifice diameter NTO / MMH, mm	1.753 / 1.338	0.791 / 1.081
Jet velocity NTO / MMH, m/s	15.7 / 23.0	15.7 / –
ΔP / injector-end pressure, NTO / MMH	20 % / 26 %	shared manifolds
Momentum ratio, spray angle	1.31, +4.4°	–

Table 13: Acoustic modes and cavities.

Quantity	Value
Chamber sound speed, m/s	1,236
1T / 1R / 1L frequency, Hz	2,984 / 6,210 / 1,947
Quarter-wave cavity depth, mm (12 cavities)	59.4
Injector mass (304L), kg	10.0

9 Feed and Pressurization System

9.1 Pressure ladder and lines

The pressure ladder (Fig. 8) starts at the chamber and adds each loss on the way back to the tank: the injector, the engine valve, a trim orifice, a filter, the gimbal flex line, the feed line, a latch valve and the tank outlet. Both tanks run at the same regulated pressure, 1.28 MPa; the trim orifices make up the difference between the two sides. Lines were sized for 4 m/s or less (Table 14); because the engine valves close in about 50 ms, the pressure surge on shutdown stays small.

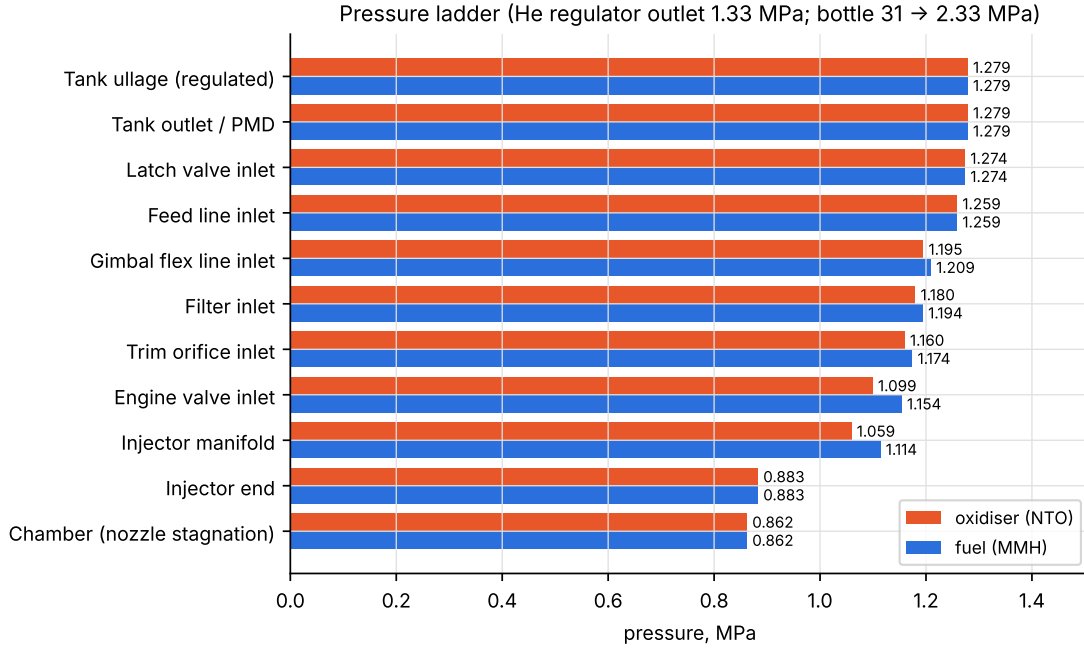


Figure 8: Pressure ladder at the design point.

Table 14: Feed lines.

Parameter	NTO	MMH
Line length (assumed), m	2.0	5.5
Tube OD $\times$ wall, in	1.5 $\times$ 0.065	1.5 $\times$ 0.065
Velocity, m/s	3.61	3.60
Line + fittings ΔP , kPa	63.9	49.6
Surge at 50 ms valve closure, MPa	0.416	0.693
Trim orifice ΔP , kPa	60	20

9.2 Tanks

Each tank holds the spreadsheet propellant mass at the spreadsheet density, plus 5 % ullage (gas space and room for the liquid to expand when warm) and 0.5 % for the PMD. One titanium sphere per propellant was kept, as in the PDR: 2.63 m (MMH, upper) and 2.64 m (NTO, lower). Stacked, they are 5.37 m tall and fit in the 6 m by 4.98 m section. The walls are 2.2 mm Ti-6Al-4V, sized with a burst factor of 2.0 on the 1.47 MPa maximum operating pressure (Table 15). A screen-channel PMD in each tank holds liquid over the outlet in weightlessness, so the engine can start without first firing thrusters to settle the propellant.

Table 15: Propellant tanks.

Parameter	MMH	NTO
Propellant mass, kg	7,925	13,075
Propellant volume (sheet density), m ³	9.057	9.080
Tank volume (+5 % ullage, +0.5 % PMD), m ³	9.555	9.580
Sphere inner diameter, m	2.633	2.635
Max. operating pressure, MPa	1.47	1.47
Wall thickness (Ti-6Al-4V), mm	2.16	2.16
Shell / PMD mass, kg	271 / 15	272 / 15

9.3 Helium pressurization

Helium pushes the propellants out of the tanks. As each tank empties, helium fills the space the liquid leaves, so at the end both tanks are full of helium at 1.28 MPa. That helium is stored at 31 MPa in four composite bottles and fed through regulators that hold the tanks at a constant pressure while the bottles run down (Fig. 9). The bottles cannot be emptied completely: the regulator needs about 1 MPa above its outlet, so the bottles stop being useful at 2.33 MPa and the helium left in them is carried too.

How much helium is needed depends on the bottle temperature. If the bottles stay warm as they empty, 44 kg is enough; if the gas cools as it expands, as it does in a fast draw, it loses pressure faster and 73 kg is needed. The design carries the cold-bottle amount at 273 K plus 10 % margin, 87 kg, in four 460 L bottles (0.96 m diameter) that fit in the gap beside the tanks (Table 16). In this cold case the bottles still hold 6.0 MPa when the tanks are empty, above the 2.33 MPa floor; that is the margin.

Table 16: Helium system.

Item	Value
Helium if the bottles stay warm (isothermal), kg	43.9
Helium if the bottles cool as they empty (adiabatic), kg	73.2
Carried (adiabatic at 273 K + 10 %), kg	86.5
Bottles	4 composite spheres, 460 L each, 0.96 m dia., 31 MPa
Bottle mass (all four), kg	380
Bottle pressure when the tanks are empty, MPa	5.96

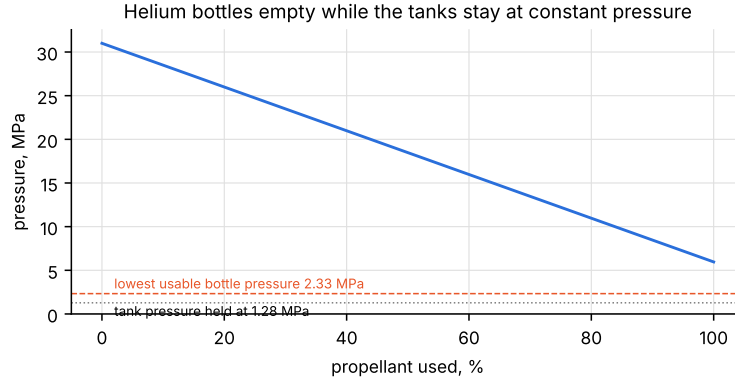


Figure 9: Helium bottle pressure as the propellant is used (design case).

10 Materials

Table 17 compares each part's operating condition with its allowable.

- **Ti-6Al-4V (tanks, chamber case).** Strong and light, compatible with both propellants, and used on the Apollo and Orion service-module tanks. Titanium can crack in NTO that contains too little nitric oxide, so the NTO must be bought to MIL-PRF-26539 with its nitric-oxide content checked.
- **Silica phenolic (chamber and throat liner).** Chosen in Sec. 7: it chars into a good insulator and does not wear away below about 1,900 K.
- **C-103 niobium with R512E silicide coating (nozzle extension).** Keeps its strength at 1,600 K; the coating stops oxidation. Used on the Apollo service-module and OME nozzle extensions.
- **304L stainless (injector) and 321 stainless (lines).** Compatible with both propellants, easy to weld, and heavy enough in the injector to soak up heat after shutdown.
- **Composite helium bottles (titanium liner, carbon wrap).** About three times lighter than all-titanium bottles; they hold only helium.
- **PTFE and PCTFE (seals, valve seats).** Resist NTO, which attacks most rubbers; usable far beyond the 283–305 K propellant range.

Table 17: Materials analysis (allowables assumed, to be verified).

Part	Material	Operating condition	Allowable	Margin
Propellant tanks	Ti-6Al-4V	448 MPa at MEOP	828 MPa yield, 895 MPa ultimate	factor 2.00 on ultimate (2.0 required)
Chamber/throat liner	silica phenolic	1806 K surface	1,900 K (wear onset)	94 K
Liner back face / case	Ti-6Al-4V case	354 K, 108 MPa	420 K bond line, 828 MPa	66 K; factor 7.7
Nozzle extension	C-103 + R512E	1389 K	1,600 K	211 K
Injector	304L stainless	399 K after shutdown	442 K (MMH boils at 1 MPa)	43 K
Feed lines	321 stainless	23 MPa (MEOP + valve-closure surge)	205 MPa yield	factor 9.0
Helium bottles	COPV (Ti liner, carbon wrap)	31 MPa	burst $\geq 1.5 \times$ MEOP (vendor rating)	vendor item
Seals and valve seats	PTFE / PCTFE	283-305 K, NTO and MMH	usable 73-533 K (PTFE)	large

11 Consistency Checks

Table 18 summarizes the checks and Table 19 the mass build-up. The engine (171 kg) is heavier than the AJ10-190 scaled to 25 kN (110 kg) because of the ablative liner and the longer extension. The propulsion system is 1,373 kg, 21.1 % of the structural mass.

Table 18: Consistency checks.

Check	Design	Requirement / spreadsheet
Vacuum thrust, N	25,000	25,000
Vacuum I_{sp} (pessimistic), s	320.8 (303.5)	≥ 280
Mass flow, kg/s	7.9434	9.1015 (at 280 s)
Burn time for 21,000 kg, s	2,643.7	2,307.3 (at 280 s)
Liner design life, s	3,305	2,644 total firing
Exit pressure, Pa	430	vacuum only
Tank stack height, m	5.37	6.0 (SM section)
Helium bottle diameter, m	0.96	1.17 (gap beside tanks)
Engine length, m	2.636	2.642
Exit diameter, m	1.457	2.0 (Fig. 9)
Propulsion-system mass, kg	1,373 (21.1 % of M_s)	6,500

Table 19: Propulsion-system mass build-up.

Item	Mass, kg
injector	10.0
valves	14.0
liner	66.4
case	10.0
extension	40.7
gimbal actuators	20.0
misc	10.0
Engine total	171
tank MMH	271.2
tank NTO	271.9
PMDs	30.0
COPVs	379.9
helium	86.5
components	40.0
lines	12.9
supports	109.3
Feed and pressurization total	1,202
Propulsion system (share of $M_s = 6,500$ kg)	1,373 (21.1 %)

12 Crew Capsule Propulsion CAD Model and Performance Verification

12.1 Overview

This section documents the Onshape CAD model of the Project ABEONA Crew Capsule service-module propulsion system and an independent check of the main-engine performance. The model contains the 25 kN NTO/MMH main engine, the two propellant tanks, four helium pressurant bottles, the propellant feed and helium pressurization lines, and reference shells for the service module, the capsule and the launch escape system (LES).

The engine geometry and the layout were generated by the main-engine design analysis in the preceding sections and were not changed, except where noted in Sec. 12.3. The tank size, envelope and LES shape follow the Preliminary Design Review (PDR) crew-capsule drawing [C1,C2]. The vehicle numbers come from the Crew Capsule tab of `Mission_Breakdown.xlsx` [C3]. The ideal performance was re-computed with RocketCEA [C4], which wraps NASA CEA [C5], and compared with the in-house equilibrium solver used above (Sec. 12.8).

12.2 Model Construction

The whole model is one parametric Onshape feature, *ABEONA engine and tanks*. It is written in FeatureScript and generated by a Python script from the analysis outputs `engine_profile.csv` and `layout.json`. The model can therefore be rebuilt from the analysis files at any time. Units are millimetres. The origin is at the service-module (SM) base on the vehicle axis, with +Z up. Figures 10 and 11 show the full model.

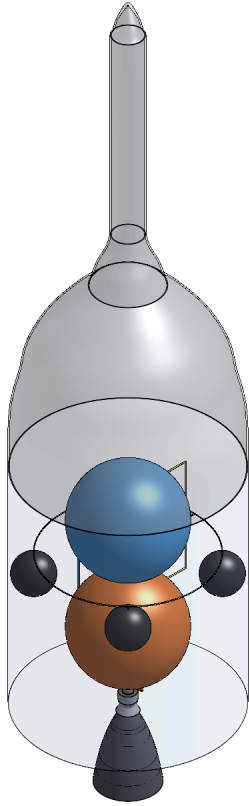


Figure 10: Full model, isometric view.

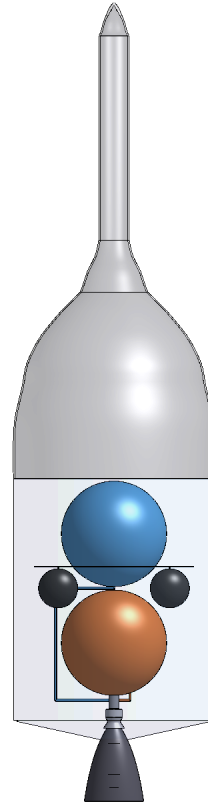


Figure 11: Full model, front view.

12.2.1 Engine

The ablative liner, the titanium case and the C-103 nozzle extension are each revolved 360° about the vehicle axis from the analysis contour. Each is a single smooth body: the gas-side and outer contours are fitted splines through the contour stations. The cylindrical chamber section is a straight line. The extension is the gas-side spline surface thickened outward by a constant 0.6 mm. It is trimmed where it meets the liner end face at area ratio 6. The liner, case and extension stay separate parts because they are different materials.

The injector is a 20 mm 304L plate at chamber diameter with twelve quarter-wave acoustic cavities, each 59.4 mm deep (Table 13). The analysis gives only the number and depth of the cavities, so they are modelled as axial $\varnothing 16$ mm bores on a 100 mm pitch radius. The valve, gimbal and head stack is a cylinder 350 mm long, from the injector face to the gimbal plane. Its density is set so that the cylinder carries the 44 kg budgeted in Table 19 for valves, gimbal actuators and miscellaneous items.

12.2.2 Tanks, pressurant and envelope

The NTO and MMH tanks are Ti-6Al-4V spherical shells. The helium bottles are spherical shells with a 20 mm wall and a density chosen to carry the composite overwrapped pressure vessel (COPV) mass in Table 19. The SM envelope (4.98 m diameter, 6 m long, 0.5 m aft cone) is a transparent reference body. The capsule and LES outer shape is a 20 mm reference shell with no mass. It was traced from the PDR drawing [C2], scaled to the 4.98 m SM diameter and the 11.8 m height from the SM top to the LES tip. The drawing's 0.41 m dimension lands on the LES nose. The tower, as drawn, is about 0.74 m in diameter, and the model follows the drawing.

12.2.3 Cut view

The feature has a *Cut view* checkbox that removes the $-Y$ half of every body. This exposes the engine section, the hollow tanks and bottles, and the feed lines, which lie in the cut plane (Figs. 12–14). Mass properties must be read with the cut view off. Onshape's own section-view tool gives the same picture without changing the model.

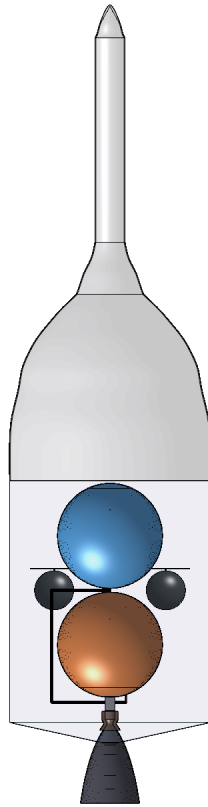


Figure 12: Cut view, front.

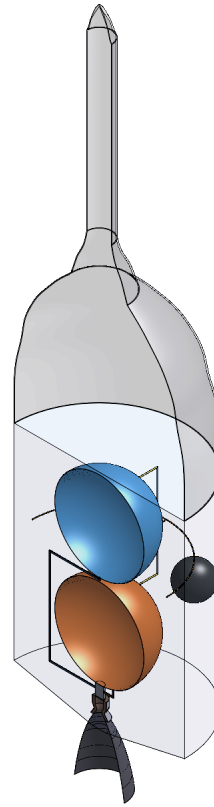


Figure 13: Cut view, isometric.

12.3 Changes from the Analysis Layout

- **Tank size.** The tanks follow the PDR drawing [C2]: 2.6 m inner diameter, 9.203 m³ each. The wall is scaled from Sec. 9.2 to 2.14 mm, which keeps the same maximum operating pressure and allowable stress. The two tanks are the same size by design. At O/F 1.65, the NTO (1,440 kg/m³) and MMH (875 kg/m³) loads occupy 9.08 and 9.06 m³ (PDR requirements 1.2.2-01/02 [C1]). Section 9.2 enlarged the tanks to 2.63 m for 5 % ullage. With 2.6 m tanks, the free volume is only 1.3 % (NTO) and 1.6 % (MMH), which is below the usual 3–5 %. This is an open item: either return to 2.63 m or offload about 3 % of the propellant.

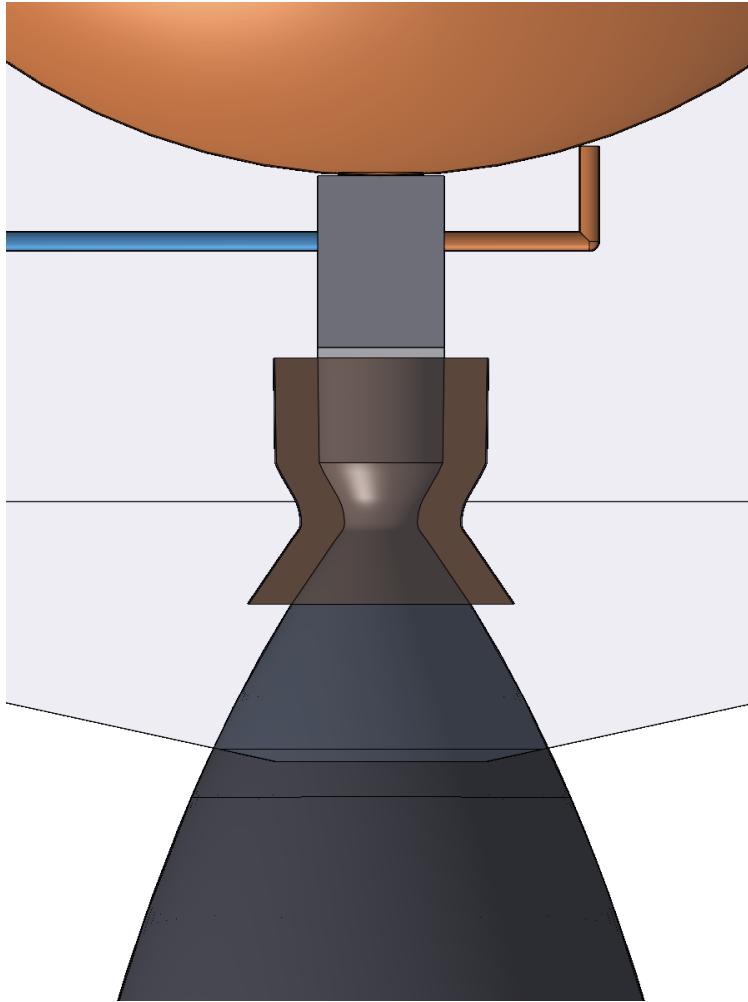


Figure 14: Cut view of the engine: head stack, injector, silica-phenolic liner and Ti case, throat, and the start of the C-103 extension, with the NTO (right) and MMH (left) feed lines entering the valve stack.

- **Tank stacking.** The analysis placed the NTO tank from its inner diameter, so its 2.16 mm wall overlapped the head stack by 19.4 cm³. Both tanks now sit on their outer surfaces. The NTO centre is at $z = 1,928$ mm and the MMH centre at $z = 4,632$ mm, with a 100 mm gap between the tanks and 65 mm clearance to the SM top. The helium bottles keep the same rules as before: 50 mm above the NTO tank and 50 mm from the SM wall.
- **Lines added.** The analysis sized the feed lines but did not lay them out. The routing and the helium line sizes are new (Sec. 12.4).

A two-bottle helium layout (2×1.206 m COPVs, same total volume) was also checked. It fitted with 376 mm tank clearance, but the four-bottle baseline was retained.

12.4 Feed and Pressurization Lines

The system is pressure-fed with regulated helium. Four 31 MPa COPVs supply a regulator that holds both tank ullages at 1.33 MPa, and the tank pressure drives the propellants to the engine (Sec. 9.2). All lines are 321 stainless steel (Table 20). The propellant line size is taken from the feed analysis (Table 14). The helium lines were sized for this model:

- **High-pressure line.** The governing load is the 31 MPa bottle pressure. A 3/8 in $\times$ 0.065 in tube then sees a 90 MPa hoop stress, a factor of 5.7 on ultimate.
- **Low-pressure lines.** These are sized by the helium flow that replaces the outflowing propellant, 3.4 L/s per tank. A 3/4 in $\times$ 0.049 in tube keeps the helium velocity near 16 m/s.

The regulator and check-valve panel is a lumped 10 kg block. Valves, filters, the flexible sections each feed line needs where it crosses the gimbal, the propellant management devices and the line supports are not modelled.

Table 20: Feed and pressurization lines.

Line	OD $\times$ wall	Basis	Route
NTO feed	1.5 in $\times$ 0.065 in	4.95 kg/s at 3.6 m/s; 64 kPa line loss	NTO lower dome ($r = 400$ mm), down, into the $+X$ side of the valve stack at $z = 500$ mm
MMH feed	1.5 in $\times$ 0.065 in	3.00 kg/s at 3.6 m/s; 50 kPa line loss	MMH bottom pole, through the tank gap, down at $x = -1,450$ mm outside the NTO tank, into the $-X$ side of the valve stack
He high pressure	0.375 in $\times$ 0.065 in	31 MPa; hoop stress 90 MPa (factor 5.7 on ultimate)	COPV top outlets to a ring manifold ($r = 1,961$ mm, $z = 3,819$ mm), then to the regulator
He low pressure	0.75 in $\times$ 0.049 in	1.33 MPa; about 16 m/s at 3.4 L/s per tank	Regulator ($+Y$, $r = 1,700$ mm) to each tank's upper dome, 60° from the top pole

12.5 Nozzle Extension Wall

The C-103 extension wall is 0.6 mm thick. Internal pressure does not size it. The gas pressure is 19.9 kPa at the attach station (area ratio 6, Mach 3.0) and 0.44 kPa at the exit. With $\sigma = pr/t$, the hoop stress is 5.7 MPa at the attach station and 0.5 MPa at the exit, which is far below the strength of C-103 at its 1,389 K peak temperature (1,600 K limit, Sec. 7).

Thickness also does not affect the wall temperature. A radiation-cooled skirt runs at radiation equilibrium, so a thicker wall only adds mass. The real sizing loads are stiffness against vibration, gimbal and handling loads, and the attach flange. The mass budget covers these with a 25 % allowance for stiffening rings, the flange and the coating (40.7 kg against 32.5 kg for the bare wall, Table 19). The CAD shows the bare wall. A taper from about 1.0 mm at the flange to 0.6 mm at the exit, with two or three hat-section stiffening rings, would represent the hardware more fully without changing the thermal design.

12.6 Mass Properties

Onshape nominal mass properties, with the cut view off, are compared with the analysis mass build-up (Table 19) in Table 21. The densities used are silica phenolic 1,750, Ti-6Al-4V 4,430, C-103 8,860, 304L 7,900 and 321 stainless 8,030 kg/m³. Items marked “lumped” have a density chosen to match the budget. The modelled masses are bare geometry. The analysis values also include flanges, rings, manifolds and tank fittings that are not drawn.

Table 21: CAD mass properties against the analysis mass build-up.

Part	Material	CAD, kg	Analysis, kg	Note
Ablative liner	silica phenolic	66.13	66.38	
Chamber case	Ti-6Al-4V	6.47	10.03	analysis adds 3 kg of flanges
Nozzle extension	C-103	33.23	40.65	analysis: 32.5 kg bare $\times$ 1.25
Injector plate	304L	6.94	10.00	analysis includes a 2.6 kg manifold
Head, valve and gimbal stack	lumped	44.00	44.00	valves, actuators, miscellaneous
NTO tank (bare shell)	Ti-6Al-4V	201.28	271.92	2.6 m ID; analysis value includes more than the shell
MMH tank (bare shell)	Ti-6Al-4V	201.28	271.22	as above
Helium COPVs ($\times 4$)	lumped	379.94	379.94	
Feed lines (NTO / MMH)	321 SS	0.70 / 8.50		
He lines (high / low pressure)	321 SS	4.17 / 2.01		
Lines total		15.38	12.93	the MMH line is 5.6 m long because it routes around the NTO tank
Regulator panel	lumped	10.00	–	part of the 40 kg components budget

12.7 Interference Check

Every pair of the 17 hardware bodies (136 pairs) was intersected on copies in Onshape. There is no interference. The largest overlap of any pair is 4×10^{-5} mm³, which is numerical noise at tangent contacts. Connected parts meet with gaps of 0–0.12 mm. The 0.12 mm gap is at the MMH outlet, where the flat tube end meets the curved dome. Selected clearances:

- NTO to MMH tank: 95.7 mm.
- Bottles to the SM wall: 50 mm.
- Bottles to the propellant tanks: 601 mm.

Against the SM envelope, only the C-103 extension lies outside (3.27 L). This is expected, because the nozzle exit is 2.01 m below the SM base and the aft cone is only 0.5 m deep.

12.8 Performance Verification with RocketCEA

The ideal design point of Table 6 was re-run with RocketCEA 1.2.3 [C4]: N₂O₄/MMH, $P_c = 0.862$ MPa, O/F 1.65, area ratio 108, infinite-area combustor, with frozen flow frozen at the chamber.

RocketCEA’s built-in propellant cards use different heats of formation from the NASA database used above. For example, N₂O₄(l) is -19.56 kJ/mol in RocketCEA against -17.55 kJ/mol in the database. The comparison therefore uses custom cards with the database values and the 293.15 K propellant temperature used above. The default cards are shown for reference (Table 22).

Every quantity agrees within 0.3 %, exit temperature included. This confirms the ideal performance on which the delivered 320.8 s specific impulse is built.

12.9 Compliance with the Mission Breakdown

The design was checked against every value in the Crew Capsule tab of the mission breakdown spreadsheet [C3] (Table 23). All values are met except the propellant volume at high propellant temperature. NTO

Table 22: Ideal vacuum performance: analysis solver against RocketCEA.

Quantity	This report	RocketCEA, same basis	Diff.	RocketCEA, default cards	Diff.
T_c , K	3,042.3	3,045.7	+0.11 %	3,044.6	+0.08 %
$\mathcal{M}_c$, g/mol	20.42	20.41	−0.03 %	20.41	−0.02 %
c^* , shifting, m/s	1,737.9	1,739.4	+0.08 %	1,738.9	+0.06 %
$C_{F,vac}$, shifting	1.9276	1.9281	+0.03 %	1.9280	+0.02 %
$I_{sp,vac}$, shifting, s	341.50	341.99	+0.14 %	341.87	+0.11 %
Exit pressure, shifting, Pa	429.9	430.3	+0.09 %	430.2	+0.06 %
c^* , frozen, m/s	1,698.6	1,699.8	+0.07 %	1,699.4	+0.05 %
$C_{F,vac}$, frozen	1.8753	1.8753	0.00 %	1.8753	0.00 %
$I_{sp,vac}$, frozen, s	324.71	325.04	+0.10 %	324.97	+0.08 %

expands as it warms. In the 2.6 m PDR tanks (9.203 m³), the 13,076 kg NTO load leaves 2.9 % free volume at 10 °C, 1.35 % at 20 °C and 0.56 % at 25 °C, and it does not fit at the 32 °C hot case (−0.57 %). The MMH tank keeps 0.8 % at 32 °C. The analysis tank sizes (2.633 m MMH, 2.635 m NTO) give 3.4 % at 32 °C and fit the SM if the gap between the tanks is reduced from 100 to 90 mm.

Table 23: Design against the Crew Capsule mission breakdown.

Item	Mission breakdown	Design
Thrust, engines	25 kN, 1 engine	25 kN vacuum, 1 engine (met)
Specific impulse	280 s	320.8 s delivered, 303.5 s pessimistic (met)
Δv	2,209 m/s	2,531 m/s, 2,394 m/s pessimistic (met)
M_0 , M_p , M_b	38,000 / 21,000 / 17,000 kg	unchanged (met)
$\dot{m} \times t_b$	9.10 kg/s $\times$ 2,307 s	7.94 kg/s $\times$ 2,644 s = 21,000 kg; liner life 3,305 s (met)
T/W at the Moon	0.405 full, 0.905 empty	unchanged (met)
Structural mass M_s	6,500 kg	propulsion system 1,373 kg, 21.1 % (met)
Propellant volume	21,000 kg at O/F 1.65	fits up to about 28 °C in 2.6 m tanks (open item)

12.10 Conclusions

The service-module propulsion system is modelled as one parametric Onshape feature built directly from the analysis outputs. It includes the engine, the PDR-size tanks, four helium bottles, sized feed and pressurization lines, and reference shells for the SM, the capsule and the LES. The model has no interferences, and its masses agree with the analysis budget once the undrawn flanges, rings and fittings are accounted for. The 0.6 mm C-103 extension wall is adequate for pressure and temperature; its sizing loads are stiffness and handling, which are covered by the ring allowance. RocketCEA reproduces the analysis performance within 0.3 %. Two items remain open: the 1.3–1.6 % ullage of the 2.6 m PDR tanks, and the flex sections the feed lines need across the gimbal.

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13 Conclusions

A pressure-fed NTO/MMH engine was designed for the Crew Capsule. At 0.862 MPa chamber pressure, O/F 1.65 and area ratio 108, it delivers 25 kN at 320.8 s vacuum I_{sp} (303.5 s pessimistic), meeting the 280 s requirement. The 80 % Rao bell (1.457 m exit) is optimized for vacuum and fits the 2.642 m length available. The silica-phenolic chamber, protected by a fuel-rich outer ring, survives 3,305 s without throat wear, and the C-103 extension stays below 1,389 K. The two titanium tanks and four helium bottles fit the PDR service-module section.

Open risks: the heat flux and the fuel-rich ring have not been calibrated by test; the ablator properties are assumed; the injector soak-back depends on the insulating ring; and the titanium tanks require controlled NTO.

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